

Engineering report template

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Description

This document intends to document blah blah blah

Standard packages

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import shutil
from scipy.constants import c, pi
from scipy import signal
from matplotlib import cm
from numpy.fft import fft, fftshift

import plotly.io as pio
import plotly.graph_objects as go
import os
import sys

# Detect if we are rendering in a headless/build environment like Quarto
# In Python, 'isinteractive' check is typically done via sys.flags
def detect_quarto_render():
    # 1. Check if we're even in a Jupyter/IPython environment
    try:
        shell = get_ipython().__class__.__name__ # type: ignore
        if shell == 'ZMQInteractiveShell':
            # 2. Check if Quarto is rendering by looking for its specific env vars
            # Quarto usually sets QUARTO_BIN_PATH or QUARTO_PROJECT_DIR during render
            is_quarto = any(key.startswith("QUARTO_") for key in os.environ)
            return is_quarto
        return False # Standard terminal script
    except NameError:
        return False # Not an interactive shell

is_rendering = detect_quarto_render()

#is_rendering = "QUARTO_ROUTES" in os.environ or not sys.flags.interactive

if is_rendering:
    # --- Light/High-Contrast Settings for Quarto HTML ---
    theme = "plotly_white"

    # Global layout defaults
```

```

layout_defaults = dict(
    template=theme,
    paper_bgcolor="white",
    plot_bgcolor="white",
    legend=dict(
        bgcolor="rgba(0, 0, 0, 0.1)",
        orientation="h", # Horizontal often fits 600x300 better
        yanchor="bottom",
        y=1.02,
        xanchor="right",
        x=1
    ),
    xaxis=dict(showgrid=True, gridcolor="rgba(0,0,0,0.12)"),
    yaxis=dict(showgrid=True, gridcolor="rgba(0,0,0,0.12)"),
    width=650,
    height=325,
    autosize=False, # Prevents the browser from stretching it
    margin=dict(l=40, r=40, t=40, b=40),
    title = dict(
        x=0.5,
        xanchor='center'
    ),
    font = dict(
        family = "Helvetica",
        size = 14
    )
)
else:
    # --- Dark/Interactive Theme ---
    theme = "plotly_dark"

    layout_defaults = dict(
        template=theme,
        paper_bgcolor="rgba(0,0,0,0)", # Transparent background
        plot_bgcolor="rgba(0,0,0,0)", # Transparent plot area
        legend=dict(
            x=0.98,
            y=0.98,
            xanchor='right',
            yanchor='top',
            bgcolor="rgba(0, 0, 0, 0.4)",
            bordercolor="rgba(0,0,0,0)"
        ),
        xaxis=dict(showgrid=True, gridcolor="rgba(255,255,255,0.12)"),
        yaxis=dict(showgrid=True, gridcolor="rgba(255,255,255,0.12)"),
        width=650,
        height=325,
        autosize=False, # Prevents the browser from stretching it
        margin=dict(l=40, r=40, t=40, b=40),
        title = dict(
            x=0.5,
            xanchor='center'
        ),
        font = dict(

```

```

        family = "Helvetica",
        size = 14
    )
)

# Apply the defaults globally
pio.templates.default = theme
pio.templates[theme].layout.update(layout_defaults); # type: ignore

# Calc demonstration packages
import handcalcs.render
handcalcs.set_option("display_precision", 2) # type: ignore
import forallpeople as si
si.environment('default', top_level= True)

# Type defining for pylance
from typing import TYPE_CHECKING
# This block is only seen by Pylance/IDE
if TYPE_CHECKING:
    m = si.m                # type: ignore
    kg = si.kg              # type: ignore
    Ohm = si.Ohm            # type: ignore
    V = si.V                # type: ignore
    A = si.A                # type: ignore
    Hz = si.Hz              # type: ignore
    W = si.W                # type: ignore
    C = si.C                # type: ignore
    F = si.F                # type: ignore
    Wb = si.Wb              # type: ignore
    T = si.T                # type: ignore
    H = si.H                # type: ignore
    Celsius = si.Celsius    # type: ignore
    lux = si.lux            # type: ignore
    s = si.s                # type: ignore
    # Add any other units you use frequently

# Some unit naming conventions
ohm = Ohm

# For debugging personal modules
# %load_ext autoreload
# %autoreload 2

print("Python stack loaded.")

```

Python stack loaded.

Topic 1

Insert text for equation here. You can number and reference equation with [Equation 1](#)

$$x = a_2$$

(1)

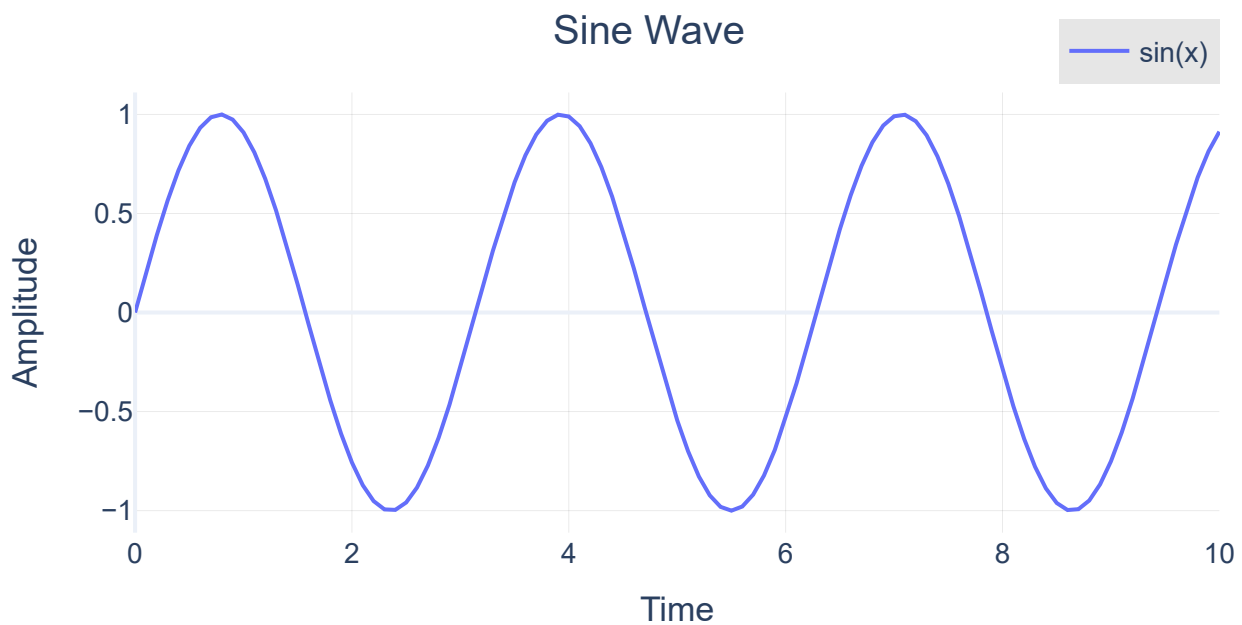
Sample plot code below

```
# 1. Data Generation
# Julia 0:0.1:10 is equivalent to np.arange
x = np.arange(0, 10.1, 0.1)
y = np.sin(2 * x)

# 2. Create the Figure
# This automatically inherits pio.templates.default
fig = go.Figure()

# 3. Add the trace (Equivalent to sp=1, alpha=0.5, and label)
fig.add_trace(go.Scatter(
    x=x,
    y=y,
    mode='lines',
    name="sin(x)",      # Equivalent to 'label'
    showlegend=True
))

# 4. Update Layout (Equivalent to title, xlabel, and ylabel)
fig.update_layout(
    title="Sine Wave",
    xaxis_title="Time",
    yaxis_title="Amplitude"
)
```



```
%%render params
# Here is an example of a Handcalcs
R_1 = 1 * ohm
I_1 = 1e3 * A
```

$$R_1 = 1.00 \, \Omega \quad I_1 = 1.00 \, \text{kA}$$

```
%%render 2
```

```
P_1 = R_1 * I_1**2
```

$$P_1 = R_1 \cdot (I_1)^2 = 1.00 \, \Omega \cdot (1.00 \, \text{kA})^2 = 1.00 \, \text{MW}$$